VIOLIN ETUDES: A COMPREHENSIVE DATASET FOR F_0 ESTIMATION AND PERFORMANCE ANALYSIS

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ABSTRACT

Violin performance analysis requires accurate and robust f_0 estimates to give feedback on the playing accuracy. Despite the recent advancements in data-driven f_0 estimators, their application to performance analysis remains a challenge due to style-specific and dataset-induced biases. In this paper, we address this problem by introducing *Violin Etudes*, a 27.8-hours violin performance dataset constructed with domain knowledge in instrument pedagogy and a novel automatic f_0-labeling paradigm. Experimental results on unseen datasets show that the CREPE f_0 estimator trained on *Violin Etudes* outperforms the widely-used pre-trained version trained on multiple manually-labeled datasets. Further preliminary findings suggest that (i) existing data-driven f_0 estimators may overfit to equal temperament, and (ii) iterative re-labeling regularized by our novel *Constrained Harmonic Resynthesis* method can simultaneously enhance datasets and f_0 estimators. Our dataset curation methodology is easily scalable to other instruments owing to the quantity of pedagogical data online. It also supports a range of MIR research directions thanks to the performance difficulty labels from educational institutions.

1. INTRODUCTION

Accurate f_0 tracking is fundamental for violin performance analysis due to the prime role of intonation in violin mastery. Musicians regard intonation and pitch accuracy as the most important criteria for assessing a string performance [1], and most of the previous work on violin performance analysis also focus on vibrato and intonation [2–5]. A study on intonation patterns of artist-level violinists [4] found that highly-regarded musicians deviate significantly from equal-temperament while remaining coherent in their intonation preferences in the close vicinity of just-noticeable difference (95% confidence intervals within just 6 cents). Another study found that violinists' intonation can be better approximated by other tuning systems, e.g., Pythagorean, rather than the standard equal-temperament [2]. From an engineering perspective, intonation analysis is reliable only if the f_0 estimates are more precise than the intonation consistency of the player. Thus, these studies imply that an f_0 estimator suitable for violin performance analysis needs to conform to a frequency precision higher than 6 cents and remain consistent irrespective of the player's deviation from equal temperament.

Alongside frequency precision, an f_0 estimator has to fulfill two more requirements for reliable performance analysis: (i) robustness to octave errors that result from the complex frequency response of the violin body and (ii) high temporal precision that can handle fast string crossings common in violin performance. However, in the current paradigm, there is a trade-off in complying with these two necessities: Most of the f_0 estimators leverage temporal post-processing stages in order to eliminate octave errors at the expense of temporal precision (e.g., Viterbi for the f_0 estimator of PRAAT [6], pYIN [7], and CREPE [8]; a custom filtering for Melodia [9]). Moreover, the Viterbi implementations of some f_0 estimators are even more restrictive: pYIN does not allow a jump bigger than 2.5 semitones between consecutive frames, and that number is 2.4 semitones for CREPE. These restrictions are detrimental to violin performance analysis, as it is common to see very fast and abrupt string crossings in violin repertoire (e.g., 21 semitone jumps in Figure 2).

Despite the above-mentioned problems, monophonic f_0 estimation is considered a mature task in MIR literature, mainly owing to the high accuracies reported in benchmark datasets. Data-driven f_0 estimators claim 96-99% accuracies in datasets such as MIR-1k [10] and MDB-stem-synth [11], which are all highly restricted in terms of performance virtuosity. Although these numbers seem promising in theory, this is not what we found when we use these f_0 estimators in the real-life analysis of advanced-level violin performances. In this paper, to better address the real-world needs of performance analysis using data-driven methods, we introduce the *Violin Etudes* [1], a 27-hour large-scale monophonic dataset comprised of pedagogical violin performances by professional violin players. We also provide our methodologies for dataset curation and automatic f_0 labeling and show the strength of our approach by outperforming the pre-trained version of CREPE f_0 estimator on its own train data.

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[1] *Violin Etudes* dataset is available for research purposes on https://doi.org/10.5281/zenodo.6564408

2. RELATED WORK

2.1 Monophonic f_0 Estimation

F_0 estimation is an essential step for many tasks in music and speech processing. Early f_0 estimators were handcrafted methods [6, 7, 12–19], but today data-driven methods [8, 20–24] are preferred following the general trend of deep learning. Both in handcrafted and data-driven techniques, f_0 estimation literature can be classified into two main approaches: time-domain and frequency-domain. Time-domain approaches used to utilize techniques such as auto-correlation during the times of handcrafted signal processing [6, 7, 12–15], while nowadays end-to-end deep learning methods directly use the time domain signal as their input [8, 20, 23, 24]. Frequency-domain approaches, on the other hand, were used to apply spectral analysis and select spectral peaks with signal processing [16–19]; but recently data-driven frequency-domain approaches require a spectrogram derivative as their input signal [21, 22]. A final important concept to mention here is self-supervised f_0 estimation, adopted by SPICE [22] and DDSP-inv [24]. Although the f_0 estimation strategy used in this paper is based on supervised learning, our combined procedure of iterative f_0-labeling is analogous to a self-supervision applied post-training.

2.2 Automatic f_0-labeling

To guarantee the f_0 label correctness for semi-automatically labeled datasets, Salamon et al. created an analysis-synthesis method [11] which forces the audio to represent any f_0 error in the annotation. The method was used for creating the resynthesized MDB-mf0-synth and Bach10-mf0-synth datasets. However, applying this method to unlabeled datasets is yet to be explored.

2.3 Large Scale Performance Datasets

Collecting large-scale datasets from community platforms has long been an important data source for research on action recognition and multimodal learning. The data collection procedure most often involves searching for keywords on YouTube, generally without collecting further metadata other than the query term itself. To our knowledge, the three main instrument performance datasets collected in this fashion are aimed at self-supervised source separation and spatial localization: MUSIC dataset [25] consists of solo and duet performances spanning over 11 instrument categories, including 53 solo violin performances. They extended the dataset for the task of video-to-audio synthesis with additional solo performances and constructed the MUSIC-Extra-Solo dataset [26] which include 213 solo violin performances, some of which include backing track. In a similar but more constrained scenario, the Solos dataset [27] is formed by searching for 13 classical instruments and the word 'audition' on YouTube and includes 66 solo violin performances in approximately 400 minutes. However, none of these datasets provide any label or metadata on the musical content: whether they are monophonic, include different renditions of the same score, the player or recording conditions, or the supposed difficulty of the performance. The only label is the instrument name.

A more constrained and informed large-scale data collection endeavor is the GiantMIDI-piano dataset [28] where the authors queried YouTube with the piano repertoire they collected from International Music Score Library Project (IMSLP) and later transcribed the audio into MIDI. By including more meta-data such as the composers, work titles, and style, the dataset is more suitable for musical analysis. However, it is also susceptible to MIDI transcription errors. The *Violin Etudes* dataset introduced in this work has similarities to this controlled approach, but in an even more restricted scenario: By collecting query words from the pedagogical repertoire, we have control over the performance difficulty. By keeping track of the performer and providing multiple renditions for the same etudes, we control the expressivity and recording conditions. Last but not least, by manually curating and removing the works, including double stops and chords, we ensure monophony and control the timbre.

3. VIOLIN ETUDES DATASET

From the 16th century onwards, music pedagogy has been creating teaching curricula that guide the students from the very early stages of their journey to the professional level. Inspired by how humans learn an instrument, we present the *Violin Etudes* dataset, the first large-scale MIR dataset rooted in instrument pedagogy. Etudes and caprices form the backbone of the traditional violin method and are defined as study pieces presenting "a technical problem or challenge in the context of a musical setting" [29]. Alongside being vital for education, these pedagogical materials have an unparalleled potential as datasets for intelligent systems, especially MIR applications. They most often come with inherent difficulty labels and are organized in human curricula, which can be used for autonomous learning. Composers often provide textual descriptions on the purpose of the study and techniques involved (e.g., Figure 3), and they are still actively researched by instrument pedagogues on their technical content e.g. [30–32]. They are most often for solo instruments, which is favorable for signal processing. And most importantly, there are hundreds of instrument teachers actively recording their reference performances of their teaching material, which supports the much-needed data for deep learning applications.

3.1 Data Collection

While previous large-scale YouTube data collection endeavors mainly focus on data for self-supervision, we opted for a more controlled approach in curating the material and stored metadata that would be used as ground truth in many topics such as expressive performance analysis and performance difficulty analysis. The individual violin methods are selected from the standard violin curriculum by a trained violinist, but interested readers are encouraged to search for *'violin (or flute/trumpet/piano...) etudes'* to see how easily they can create similar lists, e.g., [33–36].

method	n_{unique}	n_{player}	n_{Σ}	t_{Σ}
Suzuki, Vol. 1-5	40	4	158	248.8
Dancla, Op. 84	27	2	59	131.9
Wohlfahrt, Op. 45	41	6	357	458.1
Sitt, Op. 32 Vol. 1-3	34	2	60	140.1
Kayser, Op. 20	5	8	40	81.9
Mazas, Op.36	12	4	35	100.7
Dont, Op. 37	10	3	30	68.1
Kreutzer, Études	24	4	95	229.8
Fiorillo, Op. 3	13	3	34	72.1
Rode, Op.22	7	5	35	82.5
Dont, Op. 35	7	2	14	31.7
Gavinies, Matinées	6	2	8	24.6
Total	**226**	**21**	**925**	**1670.2**

Table 1. Pedagogical methods ordered in approximate difficulty. n_{unique}: number of unique monophonic studies, n_{player}: number of distinct players, n_{Σ}: recording count per method, t_{Σ}: recording duration in minutes per method.

The most effort was spent on curating a monophonic [2] subset of these works by manually going through the scores on IMSLP and discarding all the works that include even a single double stop or chord, which corresponds to manually removing 264 of 490 studies [3]. The remaining 226 monophonic pedagogical works are summarized in Table 1 in their usual order of use in the violin curriculum.

The videos of the selected works are collected from YouTube by querying method/etude names followed by manually identifying the most reliable content creators, and then searching with queries including their names. Some videos, especially in the beginner repertoire, include different studies as multiple chapters within the same recording. These chapters are split and the audio recordings are extracted from the source videos with the highest possible quality, resulting in 925 performances.

3.2 Metadata

For each recording, we provide metadata with player ID, study No., which method/etude book it belongs to, and piece-wise difficulty rankings from multiple sources. Since difficulty is a subjective term, sources sometimes disagree on the ranking of these materials, but there are some common patterns: Suzuki Vol. 1-2 are always considered the easiest of all, with Dancla, Wohlfahrt, Suzuki Vol. 3-5 being the next. Rode, Gavinies, Fiorillo, and Dont Op. 35 form the hardest cluster. Whilst difficulty grades change from source to source, the progressive ordering within each book is universally accepted, e.g. study No. 30 from a method is always considered as harder than No. 2. Hence, *Violin Etudes* can be considered as a performance difficulty analysis dataset similar to [37].

The performers in the dataset are manually confirmed to be highly-skilled violinists, mostly violin teachers creating content for their students. As shown in Figure 1, our dataset includes performances from 21 players and is also quite skewed in favor of Players 01 and 02. Although this is a limit of the dataset from one aspect, with more than 5 hours of monophonic recordings for Player02 and 7 hours for Player01, these subsets include enough data to train violin synthesizers. In Table 1, we can see that we have a total of 925 reference performances for 226 monophonic violin studies. With some exceptions, each study has multiple renditions which allow high-level music research such as expressive performance analysis (exemplified briefly in Section 5.1), or low-level audio representation learning as exemplified in the f_0 estimation experiments in Section 5.2.

Despite its many strengths, the current state of *Violin Etudes* has two main flaws. Although the scores are included, only Mazas and Kayser etudes are in a machine-readable format, and we do not have their aligned scores. As repertoire gets harder, it gets harder to find non-commercial reference recordings. Thus, recording distribution in the dataset is limited by method popularity and difficulty, e.g., Matinées in Table 1 is only 24 minutes.

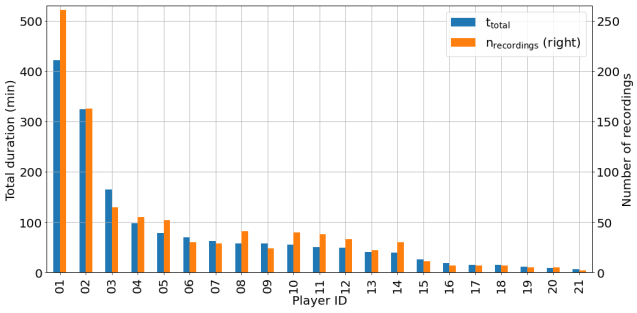


Figure 1. Total performance duration (left) and number of recordings (right) per PlayerID.

4. AUTOMATIC F_0 LABELING

The f_0 labels in the Violin Etudes dataset are automatically generated through a novel iterative f_0-labeling strategy based on two assumptions: (i) the recordings are monophonic, i.e. we can apply harmonic analysis with respect to an initial $\hat{f}_0$ estimate, and (ii) all the frames have a similar harmonic structure, i.e. violin is the sole instrument. After obtaining raw $\hat{f}_0$ estimates through a data-driven f_0 estimator, we search for harmonic peaks around the $\hat{\mathbf{f}}_{0:N}$ multiples of the $\hat{f}_0$ estimate for each frame through a modified version of Spectral Modeling Synthesis (SMS) [38]. We then force the harmonics to follow the $\hat{f}_0$ label similar to the analysis/synthesis framework of Salamon et al. [11], with additional novel constraints on instrument modeling and harmonic consistency to silence low-confidence segments. The remaining data is smaller in amount, yet it is more reliable to be used in the training of a new f_0 estimator. We then retrain the f_0 estimator using these resynthesized versions and extract new f_0' estimates, and repeat the process. Thus, as exemplified in Figure 2, both the f_0 estimator's performance and the dataset's f_0 label quality are enhanced simultaneously by interacting with one another.

[2] Here monophonic simply refers to *single note at a time*, i.e. removal of superposed notes, rather than the musicological definition.

[3] Unlike the use of violin in popular music, the classical and pedagogical violin repertoire exhaustively include double stops and chords.

4.1 Constrained Harmonic Resynthesis

First introduced by Serra et al. [38], Spectral Modeling Synthesis (SMS) has had widespread adoption in audio signal processing, including a recent revival with differential Digital Signal Processing (DDSP) [39]. We adopt the harmonic model from SMS to create constrained f_0 labels where we force the synthesis to follow the label or simply silence the frame if it does not satisfy the constraints.

4.1.1 Harmonic Analysis

Waveforms in 44.1kHz are analyzed with 1025-sample Blackman-Harris windows and hop size of 128 using `harmonicModelAnal` function from the SMS-tools [38]. The DSP-based f_0 detection algorithm of the `harmonicModelAnal` is replaced with our external $\hat{f}_0$ estimates; and harmonic amplitudes, frequencies, and phases are searched around the $\hat{\mathbf{f}}_{0:39}$ with a harmonic deviation slope of 0.001. We refer to [38] for further details.

4.1.2 Instrument-modeling Constraint (IC)

Instrument timbre defines the harmonic amplitudes $\mathbf{A}_{0:39}$ we see on top of an $\hat{f}_0$ estimate. Thus, if we know the instrument model, i.e. $P(\mathbf{A}_{0:39}|\hat{\mathbf{f}}_{0:39})$, we can assess the correctness of an estimate from harmonics. We use this idea as an additional layer of regularization of the label quality and removal of automatic-labeling artifacts. We learn an approximate model for violin resonance structure by linearly dividing the violin frequency range into N regions and fitting elliptic envelopes to the first 12 harmonic amplitudes for each of these N regions, where we experimented with $N = 50$ and 100. If this elliptic envelope model detects an anomaly in the harmonic amplitudes, we silence the frame.

4.1.3 Harmonic Consistency Constraint (HCC)

As we will show in Figure 4, an f_0 estimator is prone to overfitting problems associated with its training set distribution. To remedy this and ensure a reliable spectral distribution for our labels, we employed the Two-Way-Mismatch (TWM) procedure [19] between harmonic peaks $\hat{\mathbf{f}}_{0:39}$ and $\hat{f}_0$ candidates around the initial $\hat{f}_0$ estimate. For each voiced segment, 33 pitch candidates are selected in the range $(\hat{f}_0 - 16c, \hat{f}_0 + 16c)$ with 1 cent intervals. The candidate with the lowest Two-Way-Mismatch-error is selected to be the new $\hat{f}'_0$ estimate if its TWM-error is smaller than 5.0, and if this Harmonic Consistency Constraint is not satisfied, the frame is silenced. Finally, the constrained harmonic frequencies $\hat{\mathbf{f}}'_{\mathbf{0:39}}$ are set to the *exact multiples* of this final $\hat{f}'_0$ estimate before resynthesis.

4.1.4 Sinusoidal Synthesis

The constrained harmonics are resynthesized using the sinusoidal model from SMS [38]. Any segments shorter than 50ms are muted before synthesis to reduce artifacts. Furthermore, for each resynthesized recording, we also resynthesize its replicate with pitch shifts: both harmonics and f_0 labels are shifted with random microtonal pitch shifts in the range of 5-55 cents to ensure the statistical diversity of our labels. We found that training f_0 estimator with shifted versions of the recordings increases the stability of the estimator against equal temperament deviation.

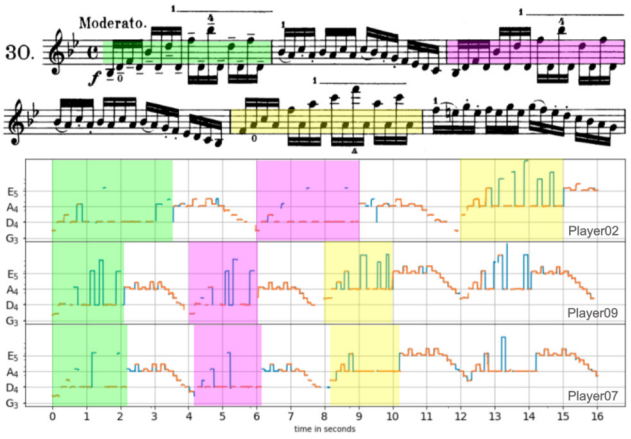


Figure 2. Iterative f_0 labeling exemplified in Kreutzer Etude No.30 performed by Players 02, 09, and 07. Constrained harmonic resynthesis acts as a barrier to wrong f_0 estimates and creates discrepancies in the initial f_0 contours (orange). Notice that most of these discrepancies are filled after the first iteration of finetuning (blue), especially in the highlighted string crossings.

4.2 Iterative Refinement of f_0 Labels

Initial f_0 tracks of the dataset are generated via the CREPE [8] convolutional f_0 estimator with 1 ms intervals and a custom Viterbi decoding to incorporate some heuristics we know about the violin repertoire. CREPE has 360 bins in its final layer with 20 cents between consecutive bin centers, i.e., states. In their implementation, they apply Viterbi with constant state observation probabilities without utilizing the confidences given by the algorithm. We replaced this by decoding with CREPE confidences as posteriors, similar to standard ANN-HMM posteriorgram decoding [40]. Using our prior knowledge of violin repertoire, we decided the Viterbi transition probabilities empirically as a weighted sum of two Gaussians to allow for fast string jumps while encouraging continuous f_0 contours: Gaussian with $\sigma_1 = 6$ semitones in Equation 1 enables fast string jumps, whilst continuous contours are encouraged by weighting the other Gaussian ($\sigma_2 = 40$ cents) with 9.

$$Pr(s_j(t+1)|s_i(t)) = \frac{1}{30\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{s_j(t+1) - s_i(t)}{30}\right)^2\right) + \frac{9}{2\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{s_j(t+1) - s_i(t)}{2}\right)^2\right) \quad (1)$$

After the initial f_0 estimates are obtained, (audio, f_0) pairs go through the Constrained Harmonic Resynthesis which silences out the wrong estimates as described in Section 4.1. The remaining (constrained audio, constrained f_0) pairs are used for finetuning of the f_0 estimator, which produces new estimates as exemplified in Figure 2. This iterative labeling process can be thought of as an Expectation-Maximization hybrid akin to ANN-HMMs [40].

	Violin		Other Inst.	
	RPA50	RPA5	RPA50	RPA5
Pretrained [8]	96.4	68.3	**96.2**	68.1
HCC	96.3	83.8	95.2	76.4
HCC + IC50	**96.7**	84.0	94.6	**76.5**
HCC + IC100	**96.7**	**84.2**	94.8	75.7
Sawtooth	68.3	49.5	56.8	37.2

Table 2. The pre-trained CREPE compared with training-from-scratch on different versions of *Violin Etudes*. Tests are conducted on the unseen URMP dataset. HCC: Harmonic Consistency Constraint. IC: Instrument-modeling Constraint. Sawtooth: Single-timbre control group.

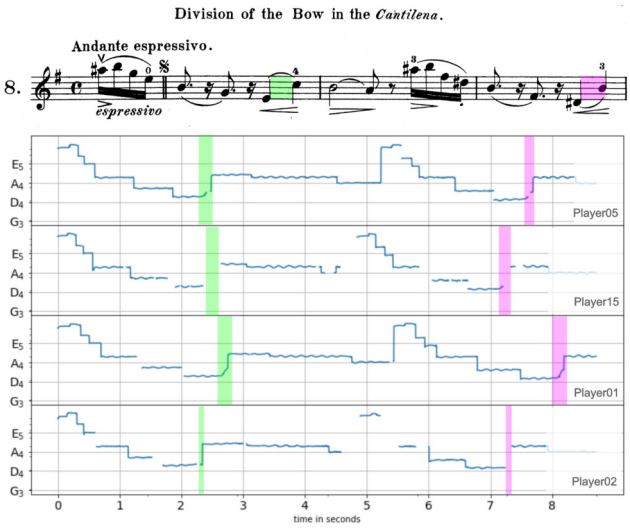


Figure 3. First line of Mazas Etude No. 8 with constrained f_0 labels of 4 performances. From the composer's words, we see that this etude targets teaching *division of the bow in the Cantilena*. f_0 tracks are automatically extracted with the method described in Section 4. Highlighted hand position changes are discussed in Section 5.1

5. EXPERIMENTS

5.1 Qualitative Performance Analysis

In this section, we give two brief examples of how *Violin Etudes* enable research on the pedagogically-motivated analysis of reference performances. In Figure 3, Mazas Op.36 No.8 with f_0 tracks of 4 different renditions can be seen: We can observe that both Players 05 and 01 have very expressive slides at the position changes indicated with green (E_4-C_5) and pink ($D_4^\sharp$-B_4), while Player02 landed into these notes directly -note that such analyses are not possible for corpora annotated with pYIN [7] or CREPE [8] since their Viterbi implementations smooth every jump. But, our method, too, has flaws: analysis of Player15 for the same section is inconclusive due to our design choice of removing unreliable automatic f_0 estimates.

Having multiple reference recordings also enable the analysis of performance tempo. In Figure 2, we see that Players 09 and 07 agree on a considerably faster pace for *Moderato* on Kreutzer Etude No.30, while Player02 play the same study a lot slower and sticks to this choice throughout the performance. Although we did not include it here, studying bowing technique analysis is also possible in *Violin Etudes*. Following the textual descriptions, players record some etudes with bowing variations. The most extreme case of this is the Player01 recording Wohlfahrt etudes with all the bowing variations, resulting in a subset of 171 recordings where some studies were repeated with up to 15 bowing variations. Moreover, the dataset allows for analyzing intonation and tuning patterns of professional violin teachers thanks to the high frequency precision of our f_0 estimates, which we will discuss in the next section.

5.2 Monophonic f_0 Estimation

Here we study the label quality of *Violin Etudes* in the context of f_0 estimator training. We train the commonly-used CREPE [8] f_0 estimator in our data, and compare it with the pre-trained version [4] trained on the following six manually-labeled and synthetic datasets: MIR-1k [10], MedleyDB [43], MDB-STEM-Synth [11, 43], Bach10-mf0-synth [11, 42], RWC-Synth [7], and NSynth [44].

We use *Violin Etudes* only for training and validation with 80/20 split, and train with a batch size of 32 and early stopping on validation accuracy. We conduct the tests on two unseen datasets where we evaluate the raw pitch accuracy (RPA, in %) with two thresholds: the conventional RPA50 quarter-tone accuracy, and our benchmark RPA5 fine-grained accuracy. Owing to the high-frequency precision requirements for performance analysis, this second metric considers the estimate accurate only if it is within 5 cents of the performance. We report frame-level accuracies without Viterbi, and separately evaluate on violin and other instruments for discussion. Since the models trained solely on violin range do not see other pitch classes, we calculate accuracies over violin range for the unseen instruments, i.e., we compute a combined accuracy from all the other instruments except violin, where the ground $f_0 \geq 190$Hz.

5.2.1 *Effect of Instrument-modeling Constraints*

In these experiments, we compare models trained on different variants of *Violin Etudes* in order to study the effectiveness of the constraints introduced in Section 4.1. Tests on the unseen URMP dataset are provided in Table 2. We see that all the three versions of *Violin Etudes* surpass the pre-trained model decisively on fine-grained accuracy RPA5 by more than 15% improvements, and increasing the number of filters from 50 to 100 in Instrument-modeling Constraint (IC) leads to marginally better results. IC also reduces the RPA50 on unseen instruments, implying that the estimator focuses excessively on the target instrument. Interestingly, one of the most conclusive results from these experiments was in the Sawtooth synthesis of the *Violin Etudes* f_0 tracks: model trained with sawtooth performs significantly better in violin compared to others, which may be due to the similarity of violin timbre to sawtooth or the effect of repertoire pitch distribution.

[4] Full CREPE model weights are taken from `https://github.com/marl/crepe/raw/models/model-full.h5.bz2`

	URMP Dataset				Bach10-mf0-synth (from train set of [8])			
	Violin		Others		Violin		Others	
	RPA50	RPA5	RPA50	RPA5	RPA50	RPA5	RPA50	RPA5
Pretrained [8]	96.4	68.3	**96.2**	68.1	98.9	89.5	99.1	**87.4**
Trained only on Violin Etudes	96.7	**84.2**	94.8	75.7	99.1	95.1	98.3	77.6
[8] finetuned on Violin Etudes	**97.0**	83.0	96.0	**77.3**	**99.2**	**95.7**	**99.2**	84.7

Table 3. Comparison of finetuning and training-from-scratch on *Violin Etudes*, tested on two instrumental datasets.

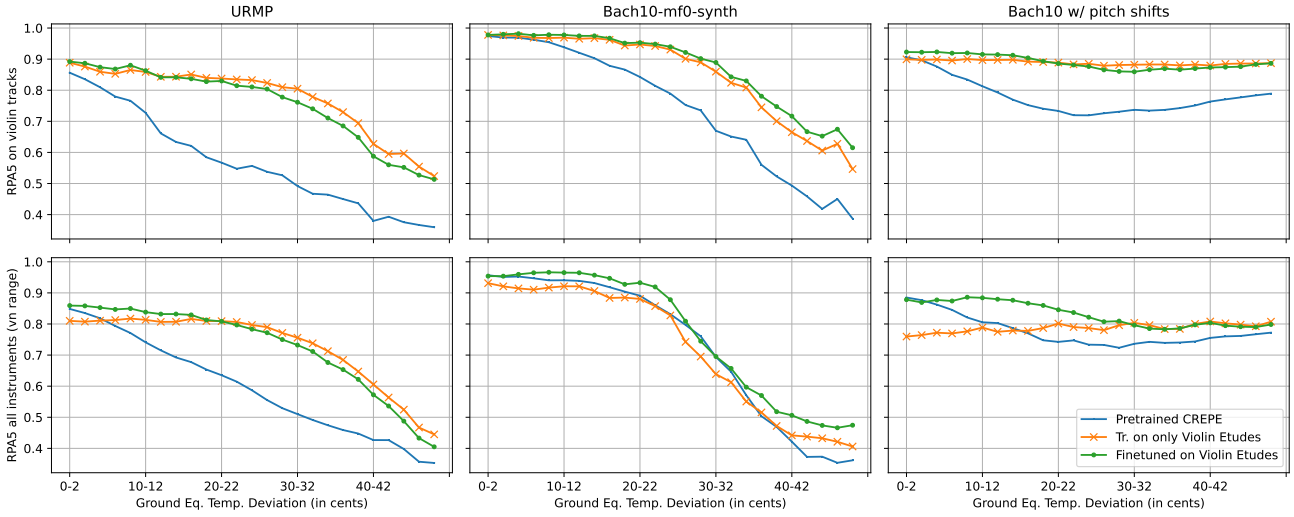


Figure 4. Fine-grained f_0 accuracy (RPA5) summarized on a grid. Music performance analysis requires precise f_0 estimates even if the player deviates from the equal temperament. However, training data-driven f_0 estimators on standard MIR datasets may induce an *auto-tuning effect*. Top: Violin-only tracks. Bottom: Over all the instruments. Left: URMP [41] dataset unseen to all the models. Center: Bach-10-mf0-synth [11, 42] from train set of [8], yet unseen to ours. Right: The conclusive experiment where we ensure the same statistics by applying microtonal pitch shifts to the Bach10 dataset.

5.2.2 *Finetuning and Generalization*

In Table 3, we study finetuning on *Violin Etudes* by testing on two datasets: alongside URMP, this time including Bach10-mf0-synth [11, 42], one of the 6 train sets of the pre-trained CREPE. We tested the estimators on the entirety of the datasets since they do not come with inherent train/test splits. We see that the finetuned model performs the best overall and remains reliable in all scenarios, including other instruments. However, interestingly, finetuning was generally more unstable and took longer to train: while the models converge after seeing around 20% of our dataset in training-from-scratch experiments, finetuning took twice as long on average. As the most interesting result of these experiments, Table 3 illustrates that our model trained on only *Violin Etudes* outperforms the pre-trained CREPE by 5.6% RPA5 on the violin tracks of its own train set.

5.2.3 *Equal Temperament and the Auto-Tuning Effect*

In Figure 4, we show that the drastic gap between the RPA5 and RPA50 performances of pre-trained CREPE can be explained by equal-temperament deviation, even on its own train data. On the other hand, our models are more robust, especially for violin-only evaluation in Figure 4. To show that this dependency is not caused by the annotation quality of the test datasets, we also experimented with microtonal pitch-shifted versions using RubberBand [5]. Yet, all our results show the importance of microtonal label distribution while training data-driven f_0 estimators. We name this phenomenon *dataset-induced auto-tuning effect*.

6. CONCLUSION

In this paper, we present *Violin Etudes*, a large-scale, comprehensive dataset for f_0 estimation and performance analysis. Our dataset curation method is easily extendable to other instruments thanks to two main novelties: 1) Reliance on pedagogy and community-driven platforms ensure abundance of material and metadata suitable for many high-level music research directions. 2) Our novel automatic f_0-labeling paradigm allows iterative refinement of labels while significantly improving the data-driven f_0 estimator as a by-product. Observing that the CREPE model converges after seeing just 20% of the *Violin Etudes* and still outperforms its pre-trained version, we argue that our dataset curation method would allow training more complex f_0 estimators that can specialize better for the needs of performance analysis. In the future, we are going to use this dataset for learning accompaniment-aware f_0 estimators and synthesizers, while further extending the *Etudes* dataset family with woodwind and brass instruments.

[5] https://breakfastquay.com/rubberband/

7. ACKNOWLEDGEMENTS

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